

Srijan Das

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EDUCATION

Ph.D., Computer Science <i>Université Côte d'Azur (Lab: INRIA, Sophia Antipolis), France</i> Supervisors: Dr. François Bremond & Dr. Monique Thonnat	2017–2020
M.Tech., Computer Science & Engineering (GPA: 9.17/10) <i>National Institute of Technology, Rourkela, India</i>	2015–2017
B.Tech., Computer Science & Engineering (GPA: 8.99/10) <i>St. Thomas' College of Engineering & Technology, Kolkata, India</i>	2011–2015

PROFESSIONAL EXPERIENCE

Assistant Professor, Department of Computer Science <i>University of North Carolina at Charlotte, USA</i>	August 2022 – Present
Postdoctoral Associate, Robotics Lab <i>Stony Brook University, USA</i> Advisor: Dr. Michael S. Ryoo	April 2021 – July 2022
Researcher, STARS Team <i>INRIA, Sophia Antipolis, France</i>	August 2017 – November 2020
Masters' Internship, STARS Team <i>INRIA, Sophia Antipolis, France</i>	January 2017 – April 2017
Teaching Assistant, Department of Computer Science <i>National Institute of Technology, Rourkela, India</i>	July 2016 – December 2016

CAREER HIGHLIGHTS

- **May 2026:** Outstanding Reviewer for CVPR 2026.
- **May 2025:** Outstanding Reviewer for CVPR 2025.
- **May 2025:** Research “Computer model aims to enhance video technology” aired on WSOC-TV ([Link](#)).
- **Feb 2025:** Secured 3rd-place in the Elderly Action Recognition (EAR) Challenge, CV4Smalls Workshop, WACV 2025.
- **Jul 2023:** Best Poster Award at MVA 2023.
- **Jun 2022:** Secured 2nd-place in Ego4D challenge under the LTA Track at CVPR 2022.
- **Oct 2019:** Granted Student Travel Award at ICCV 2019.
- **Jun 2018:** Selected in [FADEX](#) Program 2018 Edition on Artificial Intelligence.
- **Dec 2017 – Nov 2020:** Recipient of National Scholarship from UCA ED STIC.
- **Oct–Dec 2017:** Certificate of Outstanding Reviewer for FGCS and CEE Journal (Elsevier).
- **Aug 2015 – Jun 2017:** Recipient of MHRD Scholarship through GATE 2015.
- **Aug 2014:** Cognizant Certified Student.

PUBLICATIONS

Peer Reviewed Journal Publications

- [1] A. Lutheran, **S. Das**, and A. Tabarraei, “Physics-informed transformer for real-time high-fidelity topology optimization,” *Structural and Multidisciplinary Optimization* 69, 133 (2026). [doi:10.1007/s00158-026-04342-6](https://doi.org/10.1007/s00158-026-04342-6).
- [2] S. Kapse, **S. Das**, J. Zhang, R. R. Gupta, J. Saltz, D. Samaras, P. Prasanna, “Attention De-sparsification Matters: Inducing Diversity in Digital Pathology Representation Learning,” *Medical Image Analysis (MedIA)*, pp. 103070, 2024. [[Paper](#)] ([IF-10.9](#)).
- [3] R. Dai, **S. Das**, S. Sharma, L. Minciullo, L. Garattoni, F. Bremond, G. Francesca, “Toyota Smarthome Untrimmed: Real-world Untrimmed Videos for Activity Detection,” *IEEE Trans. Pattern Analysis and Machine Intelligence (TPAMI)*,

pp. 2533–2550, 2022. [\[Paper\]](#) (IF-24.3).

- [4] **S. Das**, R. Dai, D. Yang, F. Bremond, “VPN++: Rethinking Video-Pose Embeddings for Understanding Activities of Daily Living,” *IEEE Trans. Pattern Analysis and Machine Intelligence (TPAMI)*, pp. 9703–9717, 2021. [\[Paper\]](#) (IF-24.3).
- [5] **S. Das**, K. Muhammad, S. Bakshi, I. Mukherjee, P. K. Sa, A. K. Sangaiah, A. Bruno, “Lip Biometric Template Security Framework Using Spatial Steganography,” *Pattern Recognition Letters*, pp. 102–110, 2018. [\[Paper\]](#) (IF-3.9).
- [6] **S. Das**, A. Dutta, S. Sharma, S. Godbole, “A Comparative Analysis of a Novel Anomaly Detection Algorithm with Neural Networks,” *International Journal of Rough Sets and Data Analysis (IJRSDA)*, vol. 4, issue 4, pp. 1–16, 2017. [\[Paper\]](#).

Peer Reviewed Conference Publications

- [1] A. Sinha, M. S. Raj, P. Wang, A. Helmy, and **S. Das**, “MS-Temba: Multi-Scale Temporal Mamba for Efficient Temporal Action Detection,” *IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, Denver, 2026. [\[Paper\]](#). [\[acc. rate 25.4%, tier 1\]](#)
- [2] V. Belagali, S. Kapse, P. Marza, **S. Das**, Z. Li, S. Boutaj, P. Pati, S. Yellapragada, T. N. Nandi, R. K. Madduri, J. Saltz, P. Prasanna, S. Christodoulidis, M. Vakalopoulou, and D. Samaras, “TICON: A Slide-Level Tile Contextualizer for Histopathology Representation Learning,” *IEEE/CVF Conf. Computer Vision and Pattern Recognition Findings (CVPR Findings)*, Denver, 2026. [\[Paper\]](#). [\[acc. rate 10.6%, tier 1\]](#)
- [3] P. Howlader, H. Nguyen-Canh, **S. Das**, J. Xu, H. Le, and D. Samaras, “CORA: Consistency-Guided Semi-Supervised Framework for Reasoning Segmentation,” *IEEE Winter Conf. Applications of Computer Vision (WACV)*, Tucson, 2026. [\[Paper\]](#). [\[acc. rate 33.7%, tier 2\]](#)
- [4] S. Kapse, P. Pati, S. Yellapragada, **S. Das**, R. R. Gupta, J. Saltz, D. Samaras, P. Prasanna, “GECKO: Gigapixel Vision-Concept Contrastive Pretraining in Histopathology,” *IEEE/CVF Intl. Conf. Computer Vision (ICCV)*, Honolulu, 2025. [\[Paper\]](#). [\[acc. rate 24%, tier 1\]](#)
- [5] M. U. Saleem, E. Pinyoanuntapong, M. J. Patel, H. Xue, A. Helmy, **S. Das**, and P. Wang, “MaskHand: Generative Masked Modeling for Robust Hand Mesh Reconstruction in the Wild,” *IEEE/CVF Intl. Conf. Computer Vision (ICCV)*, Honolulu, 2025. [\[Paper\]](#). [\[acc. rate 24%, tier 1\]](#)
- [6] D. Reilly, R. Chakraborty, A. Sinha, M. K. Govind, P. Wang, F. Bremond, X. Le, and **S. Das**, “LLAVIDAL: A Large Language VISION Model for Daily Activities of Living,” *IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, Nashville, 2025. [\[Paper\]](#). [\[acc. rate 22.1%, tier 1\]](#)
- [7] A. Sinha, D. Reilly, F. Bremond, P. Wang, and **S. Das**, “SKI Models: Skeleton Induced Vision-Language Embeddings for Understanding Activities of Daily Living,” *39th AAAI Conf. Artificial Intelligence*, Philadelphia, 2025. [\[Paper\]](#). [\[acc. rate 23.4%, tier 1\]](#)
- [8] M. U. Saleem, E. Pinyoanuntapong, P. Wang, H. Xue, **S. Das**, and C. Chen, “GenHMR: Generative Human Mesh Recovery,” *39th AAAI Conf. Artificial Intelligence*, Philadelphia, 2025. [\[Paper\]](#). [\[acc. rate 23.4%, tier 1\]](#)
- [9] A. Lutheran, **S. Das**, and A. Tabarraei, “Latent Space Diffusion Models with Auxiliary Losses for Structural Topology Optimization,” *ASME Intl. Mechanical Engineering Congress and Exposition (IMECE)*, Vol. 8, V008T13A036, Memphis, TN, November 2025. [\[Paper\]](#). [\[tier-2\]](#)
- [10] W. Wu, C. Zheng, Z. Yang, C. Chen, **S. Das**, and A. Lu, “Frequency Guidance Matters: Skeletal Action Recognition by Frequency-Aware Mixed Transformer,” *32nd ACM Intl. Conf. Multimedia*, Melbourne, 2024. [\[Paper\]](#). [\[acc. rate 26.2%, tier 1\]](#)
- [11] E. Pinyoanuntapong, M. U. Saleem, P. Wang, M. Lee, **S. Das**, and C. Chen, “BAMM: Bidirectional Autoregressive Motion Model,” *18th European Conf. Computer Vision (ECCV)*, Milan, 2024. [\[Paper\]](#). [\[acc. rate 27.9%, tier 1\]](#)
- [12] P. Howlader, **S. Das**, H. Le, and D. Samaras, “Beyond Pixels: Semi-Supervised Semantic Segmentation with a Multi-scale Patch-based Multi-Label Classifier,” *ECCV*, Milan, 2024. [\[Paper\]](#). [\[acc. rate 27.9%, tier 1\]](#)
- [13] D. Reilly and **S. Das**, “Just Add π !: Pose Induced Video Transformers for Understanding Activities of Daily Living,” *IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, pp. 18340–18350, Seattle, 2024. [\[Paper\]](#). [\[acc. rate 23.6%, tier 1\]](#)
- [14] S. Kapse*, P. Pati*, **S. Das**, J. Zhang, C. Chen, M. Vakalopoulou, J. Saltz, D. Samaras, R. Gupta, and P. Prasanna, “SI-MIL: Taming Deep MIL for Self-Interpretability in Gigapixel Histopathology,” *CVPR*, pp. 11226–11237, Seattle, 2024. [\[Paper\]](#). [\[acc. rate 23.6%, tier 1\]](#)
- [15] A. Dutta, **S. Das**, J. Nielsen, R. Chakraborty, and M. Shah, “Multiview Aerial Visual Recognition (MAVREC): Can Multi-view Improve Aerial Visual Perception?,” *CVPR*, pp. 22678–22690, Seattle, 2024. [\[Paper\]](#). [\[acc. rate 23.6%, tier 1\]](#)

tier 1]

- [16] S. Das, T. Jain, D. Reilly, P. Balaji, S. Karmakar, S. Marjit, X. Li, A. Das, and M. S. Ryoo, "Limited Data, Unlimited Potential: A Study on ViTs Augmented by Masked Autoencoders," *IEEE/CVF Winter Conf. Applications of Computer Vision (WACV)*, pp. 6864–6874, Hawaii, 2024. [\[Paper\]](#). **[acc. rate 41%, tier 2]**
- [17] R. Dai, S. Das, M. Ryoo, F. Bremond, "Attributes-Aware Network for Temporal Action Detection," *34th British Machine Vision Conf. (BMVC)*, Aberdeen, 2023.. [\[oral presentation, tier 2\]](#)
- [18] S. Das and M. Ryoo, "Cross-modal Manifold Cutmix for Self-supervised Video Representation Learning," *18th Intl. Conf. Machine Vision Applications (MVA)*, pp. 1–6, Shizuoka, 2023. [\[Paper\]](#) **Best Poster Award**. [\[tier 3\]](#)
- [19] S. Kapse, S. Das, and P. Prasanna, "CD-Net: Histopathology Representation Learning using Pyramidal Context-Detail Network," *20th IEEE Intl. Symposium on Biomedical Imaging (ISBI)*, pp. 1–5, Cartagena, 2023. [\[Paper\]](#). [\[tier 3\]](#)
- [20] S. Das and M. Ryoo, "ViewCLR: Learning Self-supervised Video Representation for Unseen Viewpoints," *IEEE Winter Conf. Applications of Computer Vision (WACV)*, pp. 5562–5572, Hawaii, 2023. [\[Paper\]](#). **[acc. rate 35%, tier 2]**
- [21] J. Shang, S. Das, and M. S. Ryoo, "Learning Viewpoint-Agnostic Visual Representations by Recovering Tokens in 3D Space," *36th Conf. Neural Information Processing Systems (NeurIPS)*, pp. 31031–31044, New Orleans, 2022. [\[Paper\]](#). **[acc. rate 25.6%, tier 1]**
- [22] X. Li, J. Shang, S. Das, and M. S. Ryoo, "Does Self-supervised Learning Really Improve Reinforcement Learning from Pixels?," *NeurIPS*, pp. 30865–30881, New Orleans, 2022. [\[Paper\]](#). **[acc. rate 25.6%, tier 1]**
- [23] R. Dai, S. Das, K. Kahatapitiya, M. Ryoo, F. Bremond, "MS-TCT: Multi-Scale Temporal ConvTransformer for Action Detection," *IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, pp. 20041–20051, New Orleans, 2022. [\[Paper\]](#). **[acc. rate 25%, tier 1]**
- [24] R. Dai, S. Das, F. Bremond, "CTRN: Class Temporal Relational Network For Action Detection," *32nd British Machine Vision Conf. (BMVC)*, pp. 1–12, Virtual, 2021. [\[Paper\]](#). **[acc. rate 36.2%, oral presentation (top 3%), tier 2]**
- [25] R. Dai, S. Das, F. Bremond, "Learning an Augmented RGB Representation with Cross-Modal Knowledge Distillation for Action Detection," *IEEE/CVF Intl. Conf. Computer Vision (ICCV)*, pp. 13053–13064, Virtual, 2021. [\[Paper\]](#). **[acc. rate 25.9%, tier 1]**
- [26] S. Majhi, S. Das, F. Bremond, "DAM: Dissimilarity Attention Module for Weakly-supervised Video Anomaly Detection," *17th IEEE Intl. Conf. Advanced Video and Signal-based Surveillance (AVSS)*, pp. 1–8, Virtual, 2021. [\[Paper\]](#). [\[tier 3\]](#)
- [27] A. Das, S. Das, A. Dantcheva, "Demystifying Attention Mechanisms for Deep Fake Detection," *IEEE Intl. Conf. Automatic Face and Gesture Recognition (FG)*, pp. 1–7, Virtual, 2021. [\[Paper\]](#). [\[tier 3\]](#)
- [28] S. Majhi, S. Das, F. Bremond, R. Dash, P. K. Sa, "Weakly-supervised Joint Anomaly Detection and Classification," *IEEE Intl. Conf. Automatic Face and Gesture Recognition (FG)*, pp. 1–7, Virtual, 2021. [\[Paper\]](#). [\[tier 3\]](#)
- [29] R. Dai, S. Das, L. Minciullo, L. Garattoni, G. Francesca, F. Bremond, "PDAN: Pyramid Dilated Attention Network for Action Detection," *IEEE Winter Conf. Applications of Computer Vision (WACV)*, pp. 2970–2979, Virtual, 2021. [\[Paper\]](#). **[acc. rate 34.5%, tier 2]**
- [30] S. Das, S. Sharma, R. Dai, M. Thonnat, F. Bremond, "VPN: Learning Video-Pose Embedding for Activities of Daily Living," *16th European Conf. Computer Vision (ECCV)*, pp. 72–90, Virtual, 2020. [\[Paper\]](#). **[acc. rate 27.1%, tier 1]**
- [31] S. Das, M. Thonnat, F. Bremond, "Looking Deeper into Time for Activities of Daily Living Recognition," *IEEE Winter Conf. Applications of Computer Vision (WACV)*, pp. 487–496, Snowmass Village, 2020. [\[Paper\]](#). **[acc. rate 34.5%, tier 2]**
- [32] S. Das, R. Dai, M. Koperski, L. Minciullo, L. Garattoni, F. Bremond, G. Francesca, "Toyota Smarthome: Real-World Activities of Daily Living," *IEEE/CVF Intl. Conf. Computer Vision (ICCV)*, pp. 833–842, Seoul, 2019. [\[Paper\]](#). **[acc. rate 25%, tier 1]**
- [33] S. Das, A. Chaudhary, F. Bremond, M. Thonnat, "Where to Focus on for Human Action Recognition?," *IEEE Winter Conf. Applications of Computer Vision (WACV)*, pp. 71–80, Hawaii, 2019. [\[Paper\]](#). **[acc. rate 39%, tier 2]**
- [34] S. Das, M. Thonnat, K. Sakhalkar, M. Koperski, F. Bremond, G. Francesca, "A New Hybrid Architecture for Human Activity Recognition from RGB-D Videos," *25th Intl. Conf. MultiMedia Modeling (MMM)*, pp. 493–505, Thessaloniki, 2019. [\[Paper\]](#). [\[tier 3\]](#)
- [35] S. Das, K. Sakhalkar, M. Koperski, F. Bremond, "Spatio-temporal Grids for Daily Living Action Recognition," *Indian Conf. Computer Vision, Graphics and Image Processing (ICVGIP)*, pp. 1–6, Hyderabad, 2018. [\[Paper\]](#). **[acc.**

rate 33%, tier 3]

- [36] S. Das, M. Koperski, F. Bremond, G. Francesca, “Deep-Temporal LSTM for Daily Living Action Recognition,” *14th IEEE Intl. Conf. Advanced Video and Signal-Based Surveillance (AVSS)*, pp. 1–6, Auckland, 2018. [\[Paper\]](#). [\[tier 3\]](#)
- [37] S. Das, M. Koperski, F. Bremond, G. Francesca, “Action Recognition Based on a Mixture of RGB and Depth Based Skeleton,” *14th IEEE Intl. Conf. Advanced Video and Signal-Based Surveillance (AVSS)*, pp. 1–6, Lecce, 2017. [\[Paper\]](#). [\[tier 3\]](#)
- [38] S. Das, S. Sharma, S. Bakshi, I. Mukherjee, “A Framework for Pixel Intensity Modulation Based Image Steganography,” *Intl. Conf. Advanced Computing and Intelligent Engineering (ICACIE)*, pp. 3–14, Bhubaneswar, 2016. [\[Paper\]](#). [\[tier 3\]](#)
- [39] I. Mukherjee, B. Datta, R. Banerjee, S. Das, “DWT Difference Modulation Based a Novel Steganographic Algorithm,” *11th Intl. Conf. Information Systems Security (ICISS)*, pp. 573–582, Kolkata, 2015. [\[Paper\]](#). [\[tier 3\]](#)

Peer Reviewed Workshop Publications

- [1] D. Reilly, S. Das[†], and S. Das, “Learning Pose-aware Representations in Vision Transformers for Understanding Activities of Daily Living,” *8th Workshop on Affective & Behavior Analysis in-the-wild at CVPR 2025 (CVPRW)*, Nashville, 2025. [\[Paper\]](#).
- [2] M. Ali, D. Yang, A. Sinha, D. Reilly, S. Das, G. Francesca, F. Bremond, “Quo Vadis, Video Understanding with Vision-Language Foundation Models?,” *NeurIPS 2024 Workshop on Video-Language Models*, Vancouver, 2024. [\[Paper\]](#).
- [3] S. Reddy, N. Poddar, S. Das, A. Das, “Self-supervised Auxiliary Learning for Texture and Model-based Hybrid Robust and Fair Featuring in Face Analysis,” *ICPR 2024 Workshop on Fairness in Biometrics (FairBio)*, Kolkata, 2024. [\[Paper\]](#).
- [4] Y. Shi, S. Das, Y. Yan, “Upscaling Prostate Cancer MRI Images to Cell-level Resolution Using Self-supervised Learning,” *MICCAI Workshop on Computational Pathology with Multimodal Data*, Marrakesh, 2024. [\[Paper\]](#).
- [5] A. Chandra K, A. A. V., S. Das, A. Das, “Latent Flow Diffusion for Deepfake Video Generation,” *2nd Workshop and Challenge on DeepFake Analysis and Detection at CVPR 2024 (CVPRW)*, pp. 3781–3790, Seattle, 2024. [\[Paper\]](#).
- [6] P. Balaji, A. Das, S. Das, A. Dantcheva, “Attending Generalizability in Course of Deep Fake Detection by Exploring Multi-task Learning,” *Workshop on DeepFake Analysis and Detection at ICCV 2023 (ICCVW)*, pp. 475–484, Paris, 2023. [\[Paper\]](#).

Manuscripts under Review

1. A. Lutheran, S. Das, and A. Tabarraei, “Diffusion Transformers with Hybrid Conditioning for Structural Optimization.” arXiv:2605.02158 (2026). [\[Link\]](#) Under Review.
2. R. Campos, S. Maity, X. Li, S. Das, and A. Dutta, “Rethinking Attention Denoising in Vision Transformers via Negative Token Interactions.” Under Review.
3. W. Wu, Z. Guo, C. Chen, S. Das, H. Xue, P. Wang, and A. Lu, “KHMP: Frequency-Domain Kalman Refinement for High-Fidelity Human Motion Prediction.” Under Review.
4. A. Danesh Pazho, G. Alinezhad Noghre, S. Yao, B. Rahimi Ardabili, D. Reilly, S. Das, and H. Tabkhi, “ARAL RGB+D Dataset: A New Action Recognition Benchmark Dataset for Surveillance Applications.” Under Review.
5. W. Bondurant, A. Sinha, H. Le, S. Das, and S. Schuckers, “DiffSwap++: 3D Latent-Controlled Diffusion for Identity-Preserving Face Swapping.” Under Review.
6. M. K. Govind, D. Reilly, P. Wang, and S. Das, “UniLACT: Depth-Aware RGB Latent Action Learning for Vision-Language-Action Models.” Under Review.
7. D. Reilly, M. K. Govind, and S. Das, “VisCoP: Visual Probing for Domain Adaptation of Vision Language Models.” Under Review.
8. D. Reilly, M. K. Govind, and S. Das, “From My View to Yours: Ego-Augmented Learning in Large Vision Language Models for Understanding Exocentric Daily Living Activities.” Under Review.
9. A. K. Rahimian, M. K. Govind, S. Maity, D. Reilly, C. Kümmerle*, S. Das*, and A. Dutta*, “Fibottention: Inceptive Visual Representation Learning with Diverse Attention Across Heads.” Under Review. (*Equal contribution as project lead.)

Patents

1. G. Francesca, L. Minciullo, L. Garattoni, S. Das, R. Dai, F. Bremond. *Method for Recognizing Activities Using Separate Spatial and Temporal Attention Weights*. **US Patent US12586413B2** (granted). Publication No. 20230134967,

App. No. 17/754,685.

2. **S. Das** and A. Dutta. *Multiview Aerial Visual Recognition*. US Patent Application No. 63/701,106, filed September 30, 2024.
3. **S. Das** and D. Reilly. *Methods, Apparatus, and Computer Program Products for Progressively Training a Model and Generating Natural Language Outputs*. US Patent Application No. 63/693,982, filed September 12, 2024.
4. **S. Das**, R. Dai, F. Bremond, L. Minciullo, L. Garattoni, G. Francesca. *Method and System for Detecting an Action in a Video Clip*. European Patent Application EP20306343.3, filed May 2022.

EXTRAMURAL FUNDING

Peer Reviewed National and International Grants

- (PI) NSF CRII — “Understanding Activities of Daily Living in Indoor Scenarios”, 08.01.2023–01.31.2026 \$185,000, 30 months
- (PI) NSF REU Supplement — Undergraduate Research in Indoor Computer Vision, 2026 \$20,000, 12 months
- (Co-PI) NSF CITeR — “Study of Identity Preservation in Deepfake Face Swapping Tools & 3D Mask Creation”, 08.01.2024–07.31.2025, with Stephanie Schuckers (PI) \$50,000, 12 months
- (PI) The Chateaubriand Fellowship, awarded to Dominick Reilly by the French Embassy in the USA, 02.01.2024–05.31.2024, research at INRIA, Sophia Antipolis Fellowship, 4 months

Industry Grants

- (Co-PI) Lowe’s Companies, Inc. — “Smart Store Initiative”, 01.01.2023–12.31.2025, with Pu Wang (PI) \$250,000, 24 months
- (Co-PI) CorVid Technologies — “Mechanical and Thermal Design Optimization of Multifunctional Battery Systems”, 06.01.2023–05.31.2025, with Alireza Tabarraei (PI) and Anthony Bombik (PI) \$39,000 (of \$750,000), 24 months

Peer Reviewed Institutional Grants

- (Co-PI) Wake Forest CAIR AI Grant — “AI-generated Automated Physician Providers”, 12.15.2025–12.14.2026, with Steven Feldman (PI) \$3,000, 5 months
- (Co-PI) School of Data Science Summer Seed Grant 2025 — “Topology Optimization Using Physics-Informed Graph Neural Networks”, with Alireza Tabarraei (PI) \$15,000, 5 months
- (PI) UNC Faculty Research Grant 2024 — “Learning Video Representation from First-person and Third-person Views for Elderly Care”, 07.01.2024–12.31.2025 \$8,000, 18 months
- (Co-PI) School of Data Science Summer Seed Grant 2023 — “Image Deep Learning Assisted Battery Safety Risk Identification”, with Jun Xu (PI) \$5,000, 4 months

Other Peer Reviewed Resource Grants

- (PI) National Artificial Intelligence Research Resource (NAIRR) Pilot — “Developing a Large Vision-Language Model for Elderly Monitoring”, 11.08.2024–08.07.2026 \$8,500 OpenAI credits + 9,560 node hours, 18 months

STUDENT SUPERVISION

Doctoral Students Supervised

1. **Dominick Reilly** (Proposal Defense completed) August 2022 – Current
University of North Carolina at Charlotte, USA
Thesis: *Multimodal Representation Learning for Activities of Daily Living*
2. **Arkaprava Sinha** (Pre-proposal) August 2023 – Current
University of North Carolina at Charlotte, USA
Thesis: *Learning Spatio-temporal Representation for Long Untrimmed Videos*
3. **Manish Kumar Govind** (Pre-QE) January 2025 – Current
University of North Carolina at Charlotte, USA
Thesis: *Vision Language Models for Robot Learning*
4. **Weston Bondurant** (Pre-QE) January 2025 – Current
University of North Carolina at Charlotte, USA

Thesis: *3D Vision Induced Video Generation*

5. **Wenhao Chi** (Pre-QE)
University of North Carolina at Charlotte, USA
Thesis: *Egocentric Vision*

August 2025 – Current

Masters Students Supervised

1. **Manish Kumar Govind**, Research Assistant, UNC Charlotte, Aug 2023–Dec 2024.
Independent Project: Efficient Learning in Vision Transformers
2. **Vishal Bondili**, Research Assistant, UNC Charlotte, May 2023–Dec 2023.
Independent Project: Real-time Action Detection
3. **Jonathan Lorrain**, Research Assistant, UNC Charlotte, Mar 2023–Dec 2023.
Independent Project: Semi-supervised Action Recognition

Bachelors Students Supervised

1. **Smit Patel**, REU, UNC Charlotte, Jan 2026–Dec 2026.
Continual Learning of World Models
2. **Arham Hussain Inamdar**, Honors Thesis, UNC Charlotte, Jan 2026–Dec 2026.
Video Imitation Learning for Dexterous Hands
3. **Zac Howard**, Individual Study, UNC Charlotte, Jan 2026–Current.
Vision Transformer for Visual Navigation
4. **Ethan Johnson**, REU, UNC Charlotte, May 2026–Dec 2026.
World Models on Dual Arm Manipulators
5. **Nitin Chandrasekhar**, REU, UNC Charlotte, May 2025–Dec 2025.
Vision-Language-Action Models for Robot Learning
6. **Drew O'Donnell**, REU, UNC Charlotte, May 2025–Aug 2025.
Teleoperation for Robot Learning
7. **Monish Soundar Raj**, Research Assistant, UNC Charlotte, Aug 2024–May 2025.
Hierarchical Temporal Mamba for Action Detection
8. **Jack Douglass**, Independent Study, UNC Charlotte, Jan 2025–May 2025.
Mamba for Video Summarization
9. **Sindhu Gadiraju**, OUR Scholar, UNC Charlotte, Jan 2024–May 2024.
Prompt Generation for Video Conversational Models
10. **Naveen Vellaturi**, Independent Study, UNC Charlotte, Jan 2024–May 2024.
Interpretable Video Models
11. **Ian Boyle**, Independent Study, UNC Charlotte, Jan 2023–May 2023.
Video Transformers for Understanding Activities of Daily Living

Non-Degree Students Supervised

1. **Ophir Avsian**, High School Student, North Carolina School of Science and Mathematics, May 2026–August 2026.
2. **Vihaan Gaurav**, High School Student, Ardrey Kell High School, May 2026–August 2026.
3. **Srujan Ponnappalli**, High School Student, Cuthbertson High School, May 2023–Aug 2023.
4. **Sai Siddhartha Nelluri**, High School Student, Cuthbertson High School, May 2023–Aug 2023.
5. **Jacob Nielsen**, Masters' Student, SDU, Denmark, Aug 2022–Aug 2023.
6. **Tanmay Jain**, Remote Intern, UG Student, DTU, India, Aug 2022–Mar 2023.
7. **Soumyajit Karmakar**, Remote Intern, UG Student, IIIT Guwahati, India, Aug 2022–Mar 2023.
8. **Shyam Marjit**, Remote Intern, UG Student, IIIT Guwahati, India, Aug 2022–Mar 2023.

TEACHING

Major Accomplishments

- Teaching the Computer Vision course at UNC Charlotte for multiple semesters (combined UG/Graduate level). The course encourages students to pursue AI projects solving campus-related problems, including deploying real-time computer vision algorithms on GPU workstations, showcased at CS@Demos at UNC Charlotte.

- Mentored computer vision projects from external sponsors (Coral Reef Foundation, CRF) in Fall 2022 and Spring 2023; projects delivered successfully to sponsors.

Courses Taught

- **Graduate:** ITCS 5152 Computer Vision; ITCS 5051 Introduction to Computer Vision; ITCS 6010/8010 Advanced Computer Vision — UNC Charlotte, Fall 2022, Spring 2023, Fall 2023, Spring 2024, Fall 2024, Fall 2025, Spring 2026. (avg. enrollment = 26)
- **Undergraduate:** ITCS 4152 Computer Vision — UNC Charlotte, Fall 2022, Spring 2023, Fall 2023, Spring 2024, Spring 2025, Fall 2026, Spring 2026. (avg. enrollment = 20)
- **Other:** 2 Lectures at SKFGI Webinar Series 2020, “Surviving the Deep Learning Apocalypse” (August 2020).
- **Other:** 3 Lectures on [Deep Learning for Computer Vision](#) at 3IA Côte d’Azur, MSc in Data Science and Artificial Intelligence (Fall 2019).

SERVICE AND OUTREACH

External Service

Invited Talks & Presentations

- Invited Talk on “Domain Adaptation of Vision-Language Models for Health Analytics” at the Health Informatics Seminar Series, Carolina Health Informatics Program (CHIP), UNC Chapel Hill. ([Link](#))
- Invited Talk on “LLAVIDAL: A Large LAnguage VIsion Model for Daily Activities of Living” at the first special WACV 2025 Meetup Series. (May 2025)
- Invited Academic Talk on “Improved Reasoning in AI Models for Deepfake Detection” at the Martigny Biometrics Workshop, co-organised by the European Association for Biometrics (EAB), CITeR, and the Idiap Research Institute, Martigny, Switzerland. (May 2025)
- Invited research poster presentation at the Computing Community Consortium (CCC) Computing Futures Symposium, Washington, DC, USA. (May 2025)
- Panelist on “Exploring Creativity in the Age of AI: A Conversation with Head of U.S. Copyright Office” at UNC Charlotte. (April 2025)
- Guest Lecture on “Deep Neural Networks” at The University of Michigan-Dearborn. (March 2025)
- Talk on “Computer Vision Projects in CharMLab” at a RoundTable on AI, Defense Alliance of NC (DANC) and the Michael Best Law Firm. (March 2024)
- Invited Online Tech Talk on “From Pixels to Robots: Recipes for Vision-Enabled Robot Learning” at Christ University, Bangalore, India. (February 2024)
- Invited Talk on “Video Understanding using AI” at the “AI and ROS for Robotics: Theory and Practice” short-term training program at IIITDM. (December 2023)
- Invited Talk on “Computer Vision for Robot Learning” at the “AI and Machine Vision for Robotics” short-term training program at IIITDM. (June 2023)
- Talk on “From Few to More: Enhancing ViT Performance on Limited Data” at PHPC Lab, UNC Charlotte. (April 2023)
- Talk on “From Pixels to Robots: Recipes for Vision-Enabled Robot Learning” at the Seminar on Controls and Robotics, UNC Charlotte. (March 2023)
- Talk on “Quo Vadis, Computer Vision!” at the PhD Seminar, UNC Charlotte. (January 2023)
- Invited Talk on “Multiple Modalities are All You Need for Video Understanding!” at the Seventh ICCOPT, Bethlehem, Pennsylvania, USA. (July 2022)
- Invited Talk on “Multiple Modalities are All You Need for Video Understanding!” at AICTE Sponsored Short Term Course, IIITDM Kancheepuram. (March 2022)
- Talk on “Vision for Understanding Activities of Daily Living” at [SciTech Talks](#). (September 2021)
- Seminar talk on “How to Combine Modalities for Understanding Activities of Daily Living?” for CSE 600, Stony Brook University. (April 2021)
- Seminar talk on “How to Combine RGB & Poses for Understanding Activities of Daily Living?” at Université Lumière Lyon 2. (November 2020)
- Talk on “Spatio-temporal Attention Mechanisms for Activities of Daily Living” at [Nice Data Science Meetup](#). (November 2019)
- Talk on “Activity Recognition for Healthcare” at [Summer School Brain Innovation Generation @ UCA](#). (August 2018)

Journal / Conference Reviewing

- **Conferences:** ICACIE 2017, 2018; SETIT 2018; KCST 2019; ICAML 2019; AVSS 2019, 2022; WACV 2020–2026; CVPR 2021–2026; ICRA 2025; ECCV 2022, 2024; ICCV 2021, 2023, 2025; IROS 2021, 2024; ACCV 2024; NeurIPS 2023, 2024; FG 2026.
- **Journals:** TPAMI; IJCV; Pattern Recognition; CVIU (Elsevier); FGCS (Elsevier); Computer & Electrical Engineering (Elsevier); MTAP; Journal of Signal Processing: Image Communication.

Program Committees / Editorial Boards

- Serving in the Editorial Board of *Pattern Recognition* as an Associate Editor. (Feb 2026–27)
- Area Chair for NeurIPS 2025, NeurIPS 2026, BMVC 2026, BMVC 2027.
- Participation & Broadening Chair for WACV 2027.
- Associate Editor for ICRA 2024 in the Area of Vision and Sensor-Based Control.
- Senior Program Committee Member (Area Chair) for AAAI 2023, 2024, and 2026.
- Member of the DEI Committee for CVPR 2023.
- Served on NSF CISE Panel.

Community Service

- Organizer for the workshop “Artificial Intelligence for Automated Human Health-care and Monitoring” at the 17th Intl. Conf. Automatic Face and Gesture Recognition (FG) 2023.
- Session Chair for Image Understanding & Activity Recognition session at IPAS 2020.
- Mentor for B.E.N.J.I. in GirlScript Summer of Code 2019.
- Mentor for the Emerging Technology Business Incubator (ETBI), NIT Rourkela.
- Volunteer at ICACNI 2014, ICACNI 2016, ICCV 2019, ICLR 2020, ICML 2020.

Internal Service

College Committees

- CCI Research Committee Member (2025–current)
- Member, Lowe’s Endowed Position Search Committee
- Member, Machine Learning Department Chair Search Committee

Department Committees

- PhD Recruit and Review Committee Member (2022–23)
- AIRG Concentration Committee Member (2023–24)
- Research Committee Member (2023–current)

Ph.D. Dissertation / Baccalaureate (Honors) Committees

- **Chair**, Dissertation Proposal Defense Committee of Dominick Reilly (2022–current), advised by S. Das.
- Dissertation Proposal Defense Committee member for Aaron Lutheran, advised by Dr. Alireza Tabarraei (2022–current).
- Dissertation Proposal Defense Committee member for Tonmoy Hasan, advised by Dr. Razvan Bunescu (2022–current).
- Dissertation Proposal Defense Committee member for Xiangcheng Wu, advised by Dr. Xi (Sunshine) Niu (2022–current).
- Dissertation Proposal Defense Committee member for Saquib Ahmad Bhuiyan, advised by Dr. Alireza Tabarraei (2022–current).
- Dissertation Proposal Defense Committee member for Yihe Wang, advised by Dr. Xiang Zhang (2022–current).
- Dissertation Proposal Defense Committee member for Ekkasit Pinyoanuntapong, advised by Dr. Pu Wang (2022–current).
- Ph.D. Defense Committee member for Wenhan Wu, advised by Dr. Aidong Lu (2022–2025).
- Ph.D. Defense Committee member for Yaying Shi, advised by Dr. Yonghong Yan (2022–2024).
- External member of Ph.D. Defense Committee for Hoang Vu Nguyen, advised by Dr. Dimitris Samaras, Stony Brook University (2015–2025).

- External member of Ph.D. Defense Committee for Kumara Kahatapitiya, advised by Dr. Michael S. Ryoo, Stony Brook University (2019–2025).
- Graduate Faculty Representative in Ph.D. Defense Committee for Jincheng Zhang, Electrical Engineering (April 2024).
- University Honors Program Committee Member for Ishan Patel (2024).
- University Honors Program Committee Member for Arham Hussain Inamdar (2026).